

Vulnerability assessment

OpenVAS/Greenbone

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Chapter 1

Introduction

The experience will be focused on the vulnerability assessment of an IT system, using the OpenVAS/Greenbone tool. The main goal is to identify and evaluate potential security weaknesses in a target system.. This process involves scanning the system for known vulnerabilities, analyzing the results, and providing recommendations for remediation. The experience will also cover the configuration and usage of OpenVAS/Greenbone, as well as best practices for conducting vulnerability assessments effectively.

Chapter 2

Initial configuration

2.1 Greenbone

Greenbone/OpenVAS is an open-source vulnerability scanning tool that provides comprehensive security assessments for IT systems. To set up Greenbone/OpenVAS, firstly download the official Greenbone/OpenVAS OVA file from the Greenbone website.

<https://www.greenbone.net/en/openvas-free/>

Then, import the OVA file into VirtualBox 6.1 or higher. Assign at least 5GB of RAM and 2 CPU cores to the virtual machine. It's better to use about 12GB of RAM. For the first configuration, use the bridge network adapter, which allows the virtual machine to be on the same network as the host machine. Start the Greenbone/OpenVAS virtual machine and follow the on-screen instructions to complete the initial setup. Assign a user for the web interface. Once the setup is complete, access the Greenbone/OpenVAS update the feed by going to **Maintenance>Feed>Update** and wait it to synchronize (usually hours).

2.2 Debian

Debian is an open-source operating system based on GNU/Linux. It will be used as a base to create a vulnerable machine.

2.2.1 Apache

Install Apache web server and create weak certificates for it. configure Apache to allow the use of weak protocols and ciphers, such as SSLv3, TLS1.1, DES, RC4 and MD5.

```
1  #!/bin/bash
2
3  apt-get update
4  apt-get install -y apache2 php libapache2-mod-php
5  rm -rf /var/lib/apt/lists/* /var/cache/apt/archives/*
```

```

6
7 a2enmod ssl
8 a2enmod rewrite
9 a2enmod headers
10 a2enmod proxy
11 a2enmod proxy_http
12 a2dissite 000-default.conf
13 echo "ServerName 127.0.0.1" | tee /etc/apache2/conf-available/servername.conf
14 a2enconf servername
15
16 output_dir=certs
17 mkdir -p $output_dir
18 cd $output_dir
19 openssl genrsa -out weak.key 1024
20 openssl req -new -key weak.key -out weak.csr -subj
   ↪ "C=XX/ST=Lab/L=Vulnbox/O=Insecure/OU=Testing/CN=localhost"
21 openssl x509 -req -days 1 -in weak.csr -signkey weak.key -out weak.crt -sha1
22 #openssl x509 -req -days 3650 -in weak.csr -signkey weak.key -out weak_md5.crt
   ↪ -md5
23
24 mkdir -p /etc/apache2/ssl
25 cp weak.* /etc/apache2/ssl/
26
27 tee /etc/apache2/sites-available/vulnbox.conf << EOF
28 <VirtualHost *:80>
29     ServerName localhost
30
31     ProxyPreserveHost On
32     ProxyPass / http://127.0.0.1:8000/
33     ProxyPassReverse / http://127.0.0.1:8000/
34
35     # No security headers
36 </VirtualHost>
37
38 <VirtualHost *:443>
39     ServerName localhost
40
41     SSLEngine on
42     SSLCertificateFile /etc/apache2/ssl/weak.crt
43     SSLCertificateKeyFile /etc/apache2/ssl/weak.key
44
45     # Allow TLS 1.0 only if OpenSSL supports it
46     SSLProtocol all -SSLv3 -TLSv1.3 -TLSv1.2 -TLSv1.1
47     #SSLProtocol all -SSLv3
48
49     SSLCipherSuite ALL:NULL:EXPORT:LOW:MEDIUM:DES:RC4:MD5:@SECLEVEL=0
50     #SSLCipherSuite aNULL:EXPORT:LOW:ALL:@SECLEVEL=0
51     #SSLCipherSuite HIGH:!aNULL:@SECLEVEL=0
52     #SSLCipherSuite ALL:NULL:EXPORT:LOW:@SECLEVEL=0
53
54     SSLHonorCipherOrder off
55
56     #SSLCompression on
57     SSLSessionTickets On
58
59     SSLOpenSSLConfCmd MinProtocol SSLv3
60     SSLOpenSSLConfCmd CipherString DEFAULT:@SECLEVEL=0
61
62     ProxyPreserveHost On

```

```

63     ProxyPass / http://127.0.0.1:8000/
64     ProxyPassReverse / http://127.0.0.1:8000/
65
66     Header always unset Strict-Transport-Security
67     Header always unset X-Frame-Options
68     Header always unset X-Content-Type-Options
69 </VirtualHost>
70 EOF
71
72 a2ensite vulnbox.conf
73
74 tee /etc/php/7.4/apache2/php.ini << EOF
75 display_errors = On
76 display_startup_errors = On
77 allow_url_fopen = On
78 allow_url_include = On
79 disable_functions =
80 open_basedir =
81 expose_php = On
82 EOF
83
84 systemctl restart apache2

```

Any website can be used as exposed service. In our case we used an old project of ours, properly modified to be vulnerable.

```
git clone -b vulnerable https://github.com/ozneroL541/artigianato-online.git
```

To run it execute the following commands in the terminal:

```

1 cd artigianato-online/src
2 docker compose up --remove-orphans -d

```

2.2.2 Cups

Install the CUPS printing system and configure it to allow remote administration without authentication.

```

1 #!/bin/bash
2
3 apt-get update
4 apt-get install -y cups
5 rm -rf /var/lib/apt/lists/* /var/cache/apt/archives/*
6
7 tee /etc/cups/cupsd.conf << EOF
8 Listen 0.0.0.0:631
9
10 <Location />
11     Order allow,deny
12     Allow all
13 </Location>
14
15 <Location /admin>
16     Order allow,deny
17     Allow all
18 </Location>
19
20 WebInterface Yes
21 DefaultAuthType Basic

```

```
22 EOF
23
24 systemctl enable --now cups
```